

Gym App V1

New App (with Validation)

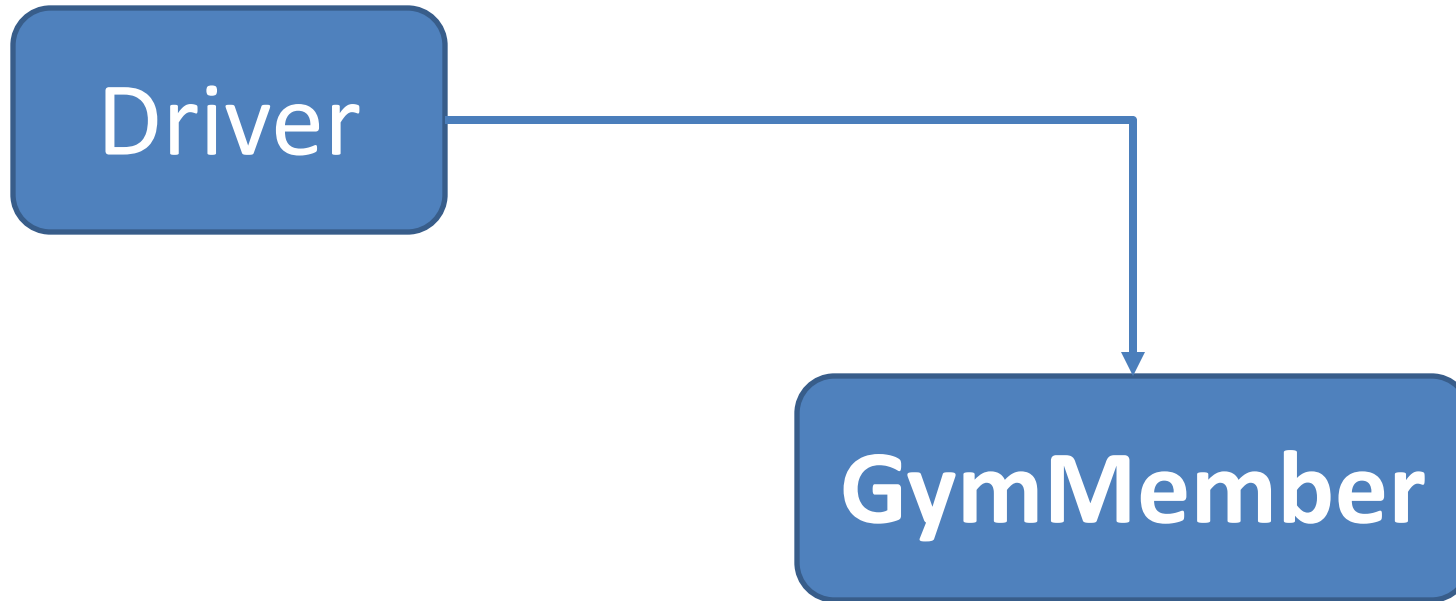
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Ms. Mairead Meagher,
Ms. Siobhán Roche.



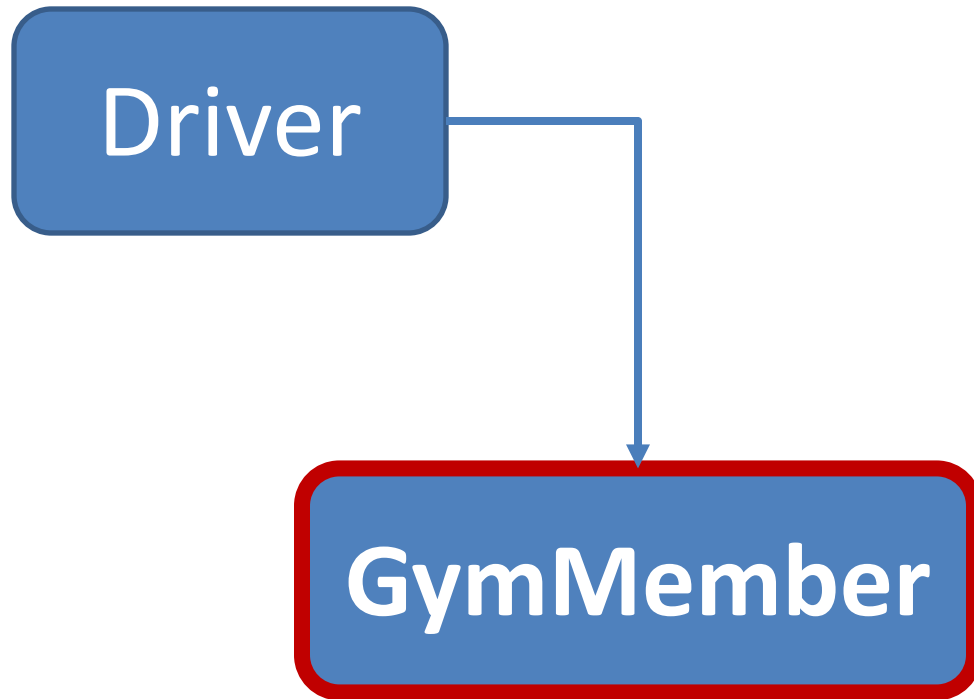
GYM APP

Version 1.0

Gym App V1.0



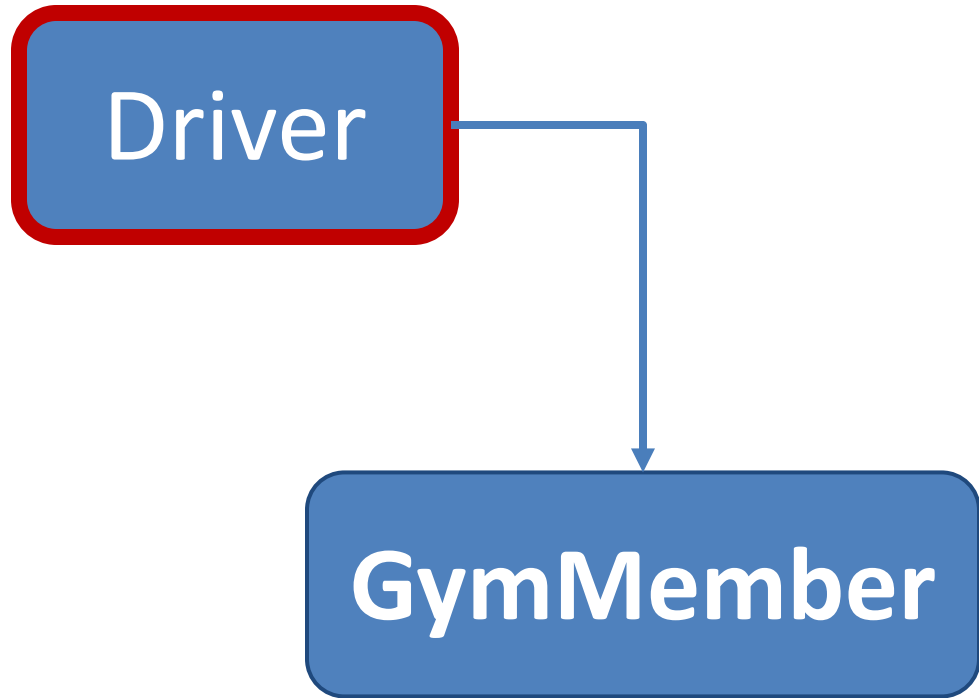
Gym App V1.0



The **GymMember** class stores **details** about a single member of a gym i.e.:

- name
- height
- weight
- membershipNumber
- if they are current members or not? (i.e. have they paid their membership fee)

Gym App V1.0



The **Driver** class:

- has the **main()** method.
- **reads** the gym member details from the user (via the console)
- **creates** a new GymMember object.
- **prints** the gym member details from the object (to the console)

Gym App V1.0 – Sample I/O



```
Run: Driver x
/Library/Java/JavaVirtualMachines/jdk-18.0.1.1.jdk/Contents/Home/b
Entering details
-----
Enter your name:          Joe Soap
Enter your height (meters): 1.7
Enter your weight (kgs):   75
Enter the membership number: 10001
Is current member (y/n):   y

Printing details
-----
Joe Soap: 1.7M, 75.0KG (Member Num: 10001, current member: true)

Process finished with exit code 0
|
```



ENCAPSULATED CLASS:

GymMember

A GymMember Class...



Object Type/ **Class** Name
i.e. GymMember

The **C** icon means it is a **Class**.

The open padlock means it is **public**.

```
▼ (C) GymMember
  (m) (🔓) GymMember(String, double, double, int, boolean)
  (m) (🔓) getName(): String
  (m) (🔓) setName(String): void
  (m) (🔓) initName(String): void
  (m) (🔓) getHeight(): double
  (m) (🔓) setHeight(double): void
  (m) (🔓) getWeight(): double
  (m) (🔓) setWeight(double): void
  (m) (🔓) getMembershipNumber(): int
  (m) (🔓) setMembershipNumber(int): void
  (m) (🔓) isCurrentGymMember(): boolean
  (m) (🔓) setCurrentGymMember(boolean): void
  (m) (🔓) toString(): String
  (f) (🔒) name: String = "Unknown"
  (f) (🔒) height: double = 0.0
  (f) (🔒) weight: double = 0.0
  (f) (🔒) membershipNumber: int = 99999
  (f) (🔒) isCurrentGymMember: boolean = false
```


A GymMember Class...



```
✓ © GymMember
  (m) GymMember(String, double, double, int, bo
  (m) getName(): String
  (m) setName(String): void
  (m) initName(String): void
  (m) getHeight(): double
  (m) setHeight(double): void
  (m) getWeight(): double
  (m) setWeight(double): void
  (m) getMembershipNumber(): int
  (m) setMembershipNumber(int): void
  (m) isCurrentGymMember(): boolean
  (m) setCurrentGymMember(boolean): void
  (m) toString(): String
```

The **f** icon means it is a **field**.

Fields
i.e. the **attributes / properties**
of the class

```
(f) name: String = "Unknown"
(f) height: double = 0.0
(f) weight: double = 0.0
(f) membershipNumber: int = 99999
(f) isCurrentGymMember: boolean = false
```

A GymMember Class...



The closed padlock means it is **private**.

The **f** icon means it is a **field**.

Fields
i.e. the **attributes / properties**
of the class

```
✓ (c) GymMember
  (m) GymMember(String, double, double, int, boolean)
  (m) getName(): String
  (m) setName(String): void
  (m) initName(String): void
  (m) getHeight(): double
  (m) setHeight(double): void
  (m) getWeight(): double
  (m) setWeight(double): void
  (m) getMembershipNumber(): int
  (m) setMembershipNumber(int): void
  (m) isCurrentGymMember(): boolean
  (m) setCurrentGymMember(boolean): void
  (m) toString(): String
  (f) name: String = "Unknown"
  (f) height: double = 0.0
  (f) weight: double = 0.0
  (f) membershipNumber: int = 99999
  (f) isCurrentGymMember: boolean = false
```

A GymMember Class...



field **type**

field **name**

The closed padlock means it is **private**.

The **f** icon means it is a **field**.

Fields
i.e. the **attributes / properties**
of the class

✓ GymMember

GymMember(String, double, double, int, boolean)

getName(): String

setName(String): void

initName(String): void

getHeight(): double

setHeight(double): void

getWeight(): double

setWeight(double): void

getMembershipNumber(): int

setMembershipNumber(int): void

isCurrentGymMember(): boolean

setCurrentGymMember(boolean): void

toString(): String

name: String = "Unknown"

height: double = 0.0

weight: double = 0.0

membershipNumber: int = 99999

isCurrentGymMember: boolean = false

A GymMember Class...Constructor



Constructor

i.e. for building objects.

▼ c GymMember
m GymMember(String, double, double, int, boolean)

The **m** icon means it is a **method**.

The open padlock means it is **public**.

Constructors have same name as the class

A GymMember Class...Constructor



Constructor

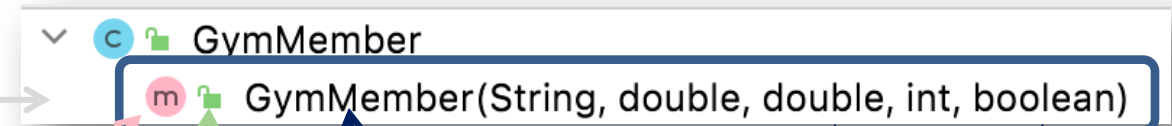
i.e. for building objects.

The **m** icon means it is a **method**.

The open padlock means it is **public**.

Constructors have same name as the class

Five **parameters**;
one for each field.



A GymMember Class...Fields and Constructor



```
private String name = "Unknown";  
private double height = 0.0;  
private double weight = 0.0;  
private int membershipNumber = 99999;  
private boolean isCurrentGymMember = false;
```

```
public GymMember(String name, double height, double weight, int membershipNumber, boolean isCurrentGymMember) {  
    initName(name);  
    setHeight(height);  
    setWeight(weight);  
    setMembershipNumber(membershipNumber);  
    setCurrentGymMember(isCurrentGymMember);  
}
```

A GymMember Class...methods



The open padlock means it is **public**.

The **m** icon means it is a **method**.

Methods
i.e. the **behaviours** of the class

```
✓ © GymMember
  (m) GymMember(String, double, double, int, boolean)
  (m) getName(): String
  (m) setName(String): void
  (m) initName(String): void
  (m) getHeight(): double
  (m) setHeight(double): void
  (m) getWeight(): double
  (m) setWeight(double): void
  (m) getMembershipNumber(): int
  (m) setMembershipNumber(int): void
  (m) isCurrentGymMember(): boolean
  (m) setCurrentGymMember(boolean): void
  (m) toString(): String
  (f) name: String = "Unknown"
  (f) height: double = 0.0
  (f) weight: double = 0.0
  (f) membershipNumber: int = 99999
  (f) isCurrentGymMember: boolean = false
```


A GymMember Class...methods



Return type

Method name

✓ GymMember

Ⓜ GymMember(String, double, double, int, boolean)

Ⓜ getName(): String

Ⓜ setName(String): void

Ⓜ initName(String): void

Ⓜ getHeight(): double

Ⓜ setHeight(double): void

Ⓜ getWeight(): double

Ⓜ setWeight(double): void

Ⓜ getMembershipNumber(): int

Ⓜ setMembershipNumber(int): void

Ⓜ isCurrentGymMember(): boolean

Ⓜ setCurrentGymMember(boolean): void

Ⓜ toString(): String

ⓕ name: String = "Unknown"

ⓕ height: double = 0.0

ⓕ weight: double = 0.0

ⓕ membershipNumber: int = 99999

ⓕ isCurrentGymMember: boolean = false

A GymMember Class...getters



getters

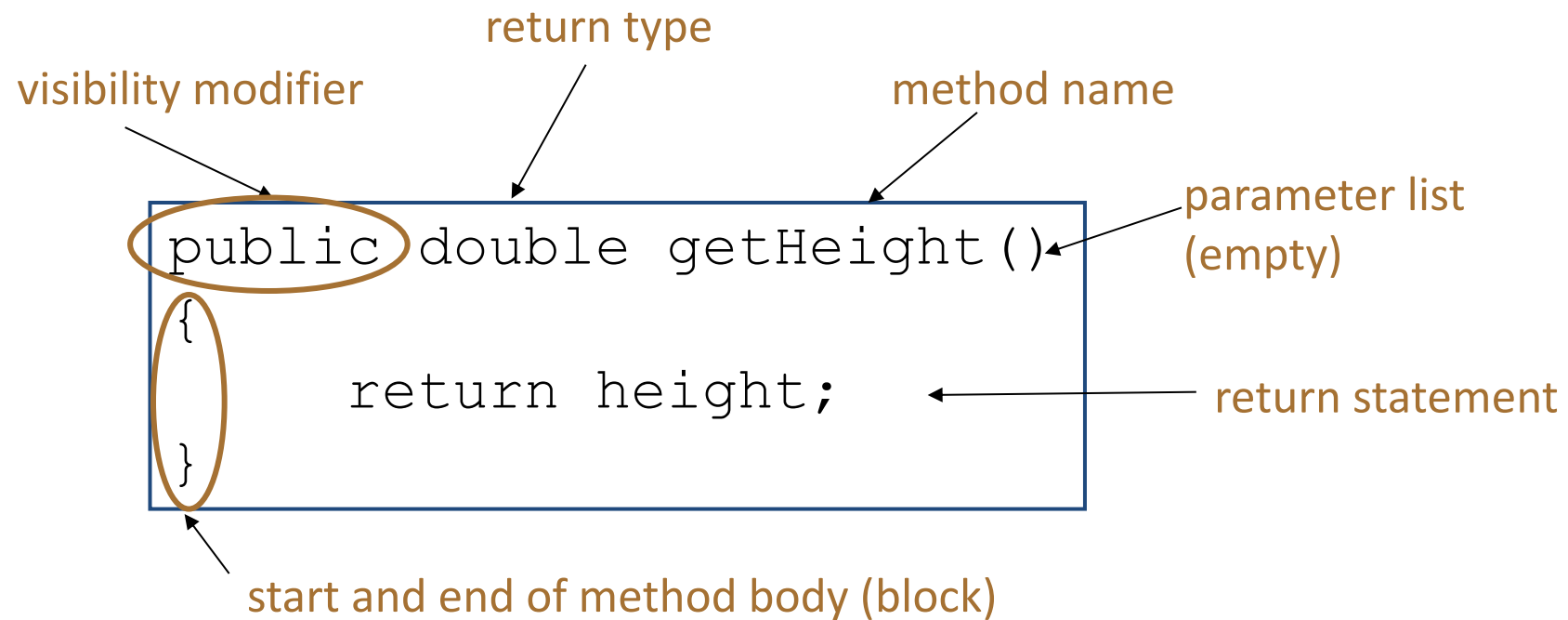
Note the different naming standard for a boolean field getter

```
✓ ⓘ GymMember
  ⓘ GymMember(String, double, double, int, boolean)
  ⓘ getName(): String
  ⓘ setName(String): void
  ⓘ initName(String): void
  ⓘ getHeight(): double
  ⓘ setHeight(double): void
  ⓘ getWeight(): double
  ⓘ setWeight(double): void
  ⓘ getMembershipNumber(): int
  ⓘ setMembershipNumber(int): void
  ⓘ isCurrentGymMember(): boolean
  ⓘ setCurrentGymMember(boolean): void
  ⓘ toString(): String
  ⓘ name: String = "Unknown"
  ⓘ height: double = 0.0
  ⓘ weight: double = 0.0
  ⓘ membershipNumber: int = 99999
  ⓘ isCurrentGymMember: boolean = false
```

Getters (Accessor Methods)

- **Accessor** methods
 - return information about the **state** of an object
 - i.e. the values stored in the fields.
- A **'getter'** method
 - is a specific type of **accessor** method and typically:
 - contains a **return statement** (as the last executable statement in the method).
 - defines a **return type**.
 - **does NOT change the object state**.

Getters



A GymMember Class...getters



```
public String getName() {  
    return name;  
}
```

```
public double getHeight() {  
    return height;  
}
```

```
public double getWeight() {  
    return weight;  
}
```

```
public int getMembershipNumber() {  
    return membershipNumber;  
}
```

```
public boolean isCurrentGymMember() {  
    return isCurrentGymMember;  
}
```

A GymMember Class...setters



setters

```
▼ GymMember
  GymMember(String, double, double, int, boolean)
  getName(): String
  setName(String): void
  initName(String): void
  getHeight(): double
  setHeight(double): void
  getWeight(): double
  setWeight(double): void
  getMembershipNumber(): int
  setMembershipNumber(int): void
  isCurrentGymMember(): boolean
  setCurrentGymMember(boolean): void
  toString(): String
  name: String = "Unknown"
  height: double = 0.0
  weight: double = 0.0
  membershipNumber: int = 99999
  isCurrentGymMember: boolean = false
```

Setters (Mutator methods)

- **Mutator** methods
 - change (i.e. mutate!) an object's state.
- A **'setter'** method
 - is a specific type of **mutator** method and typically:
 - contains an **assignment statement**
 - takes in a **parameter**
 - **changes the object state.**

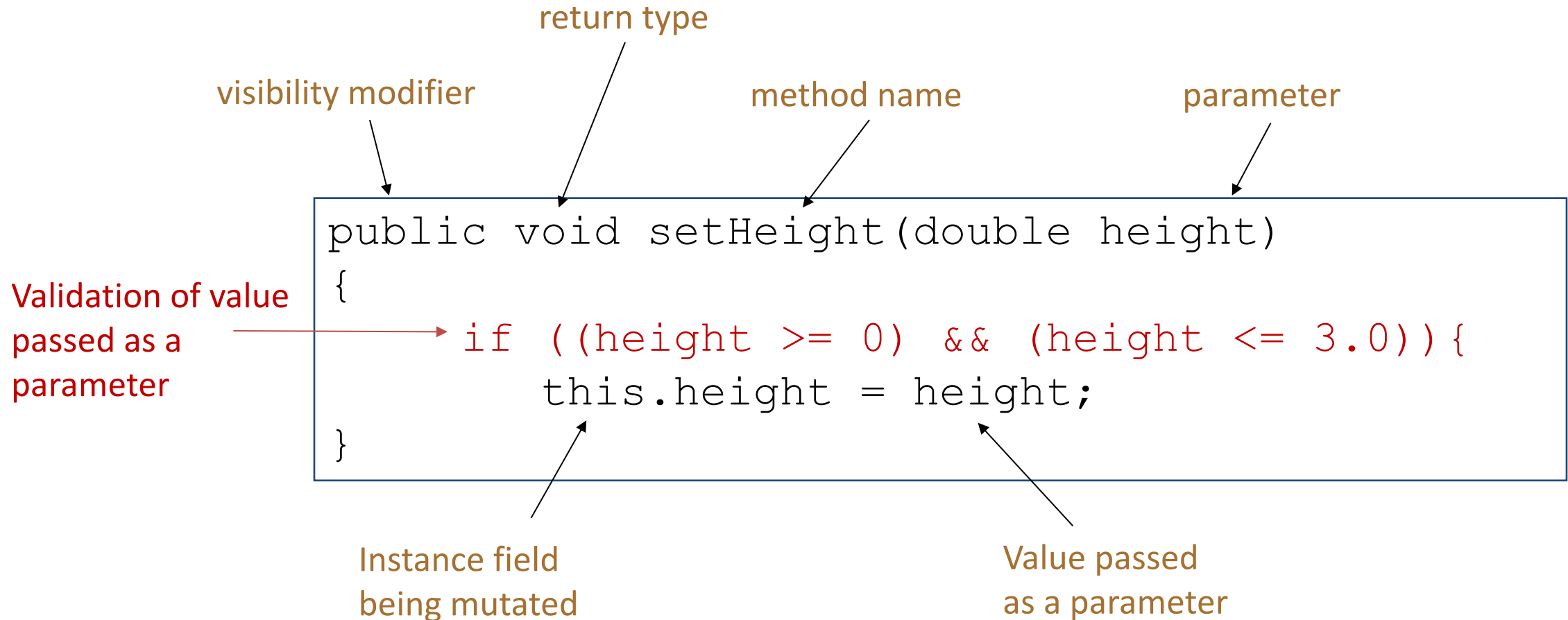
Setters – without validation

The diagram illustrates the components of a Java setter method. The code is enclosed in a blue rectangular box. Annotations with arrows point to specific parts of the code:

- visibility modifier**: Points to the word `public`.
- return type**: Points to the word `void`.
- method name**: Points to the text `setHeight`.
- parameter**: Points to the text `double height` inside the parentheses.
- Instance field being mutated**: Points to the text `this.height` on the assignment line.
- assignment statement**: Points to the equals sign `=` on the assignment line.
- Value passed as a parameter**: Points to the text `height` on the assignment line.

```
public void setHeight(double height)
{
    this.height = height;
}
```

Setters – with validation



A **GymMember** Class...special case for Strings in Constructor



```
public void initName(String name) {  
    if (name != null) {  
        if (name.length() > 30)  
            this.name = name.substring(0, 30);  
        else  
            this.name = name;  
    }  
}
```

This is used in the constructor instead of the setter (setName()). This is because we assume that the name is somewhat correct (just too long)

We will always use this technique for Strings

A GymMember Class...setters



```
public void setName(String name) {  
    if (name != null) {  
        if (name.length() > 30)  
            this.name = name.substring(0, 30);  
    }  
}
```

A GymMember Class...setters



```
public void setHeight(double height) {  
    if ((height >= 0) && (height <= 3.0)) {  
        this.height = height;  
    }  
}
```

```
public void setWeight(double weight) {  
    if ((weight >= 0) && (weight <= 500)) {  
        this.weight = weight;  
    }  
}
```

A GymMember Class...setters



```
public void setMembershipNumber(int membershipNumber) {  
    if ((membershipNumber > 0) && (membershipNumber < 99999)) {  
        this.membershipNumber = membershipNumber;  
    }  
}
```

```
public void setCurrentGymMember(boolean currentGymMember) {  
    isCurrentGymMember = currentGymMember;  
}
```

Getters/Setters

For **each instance field** in a class, you are normally asked to write:

- A **getter**
 - Return statement
- A **setter**
 - Assignment statement

A GymMember Class...toString



toString():

Builds and returns a String containing a user-friendly representation of the object state.

GymMember

GymMember(String, double, double, int, boolean)

getName(): String

setName(String): void

initName(String): void

getHeight(): double

setHeight(double): void

getWeight(): double

setWeight(double): void

getMembershipNumber(): int

setMembershipNumber(int): void

isCurrentGymMember(): boolean

setCurrentGymMember(boolean): void

toString(): String

name: String = "Unknown"

height: double = 0.0

weight: double = 0.0

membershipNumber: int = 99999

isCurrentGymMember: boolean = false

A GymMember Class...toString



```
@Override
public String toString() {
    return name + ": " +
        height + "M, " +
        weight + "KG (Member Num: " +
        membershipNumber + ", current member: " +
        isCurrentGymMember + ")";
}
```

Sample Console Output if we don't have a toString for GymMember:

GymMember@77459877

toString()

- This is a useful method and you will write a **toString()** method for most of your classes.
- **When you print an object, Java automatically calls the toString() method e.g.**

```
GymMember gymMember = new GymMember();
```

```
//both of these lines of code do the same thing
```

```
System.out.println(gymMember);
```

```
System.out.println(gymMember.toString());
```




ENCAPSULATION RECAP

GymMember

Encapsulation in Java – steps 1-3

Encapsulation Step	Approach in Java
1. Wrap the data (fields) and code acting on the data (methods) together as single unit.	<pre>public class <i>ClassName</i> { <i>Fields</i> <i>Constructors</i> <i>Methods</i> }</pre>
2. Hide the fields from other classes.	Declare the fields of a class as <u>private</u>.
3. Access the fields only through the methods of their current class.	Provide <u>public</u> setter and getter methods to modify and view the fields values.

A GymMember Class... An Encapsulated Class

1. GymMember class
wraps the data (fields)
and code acting on the
data (methods)
together as **single unit**.

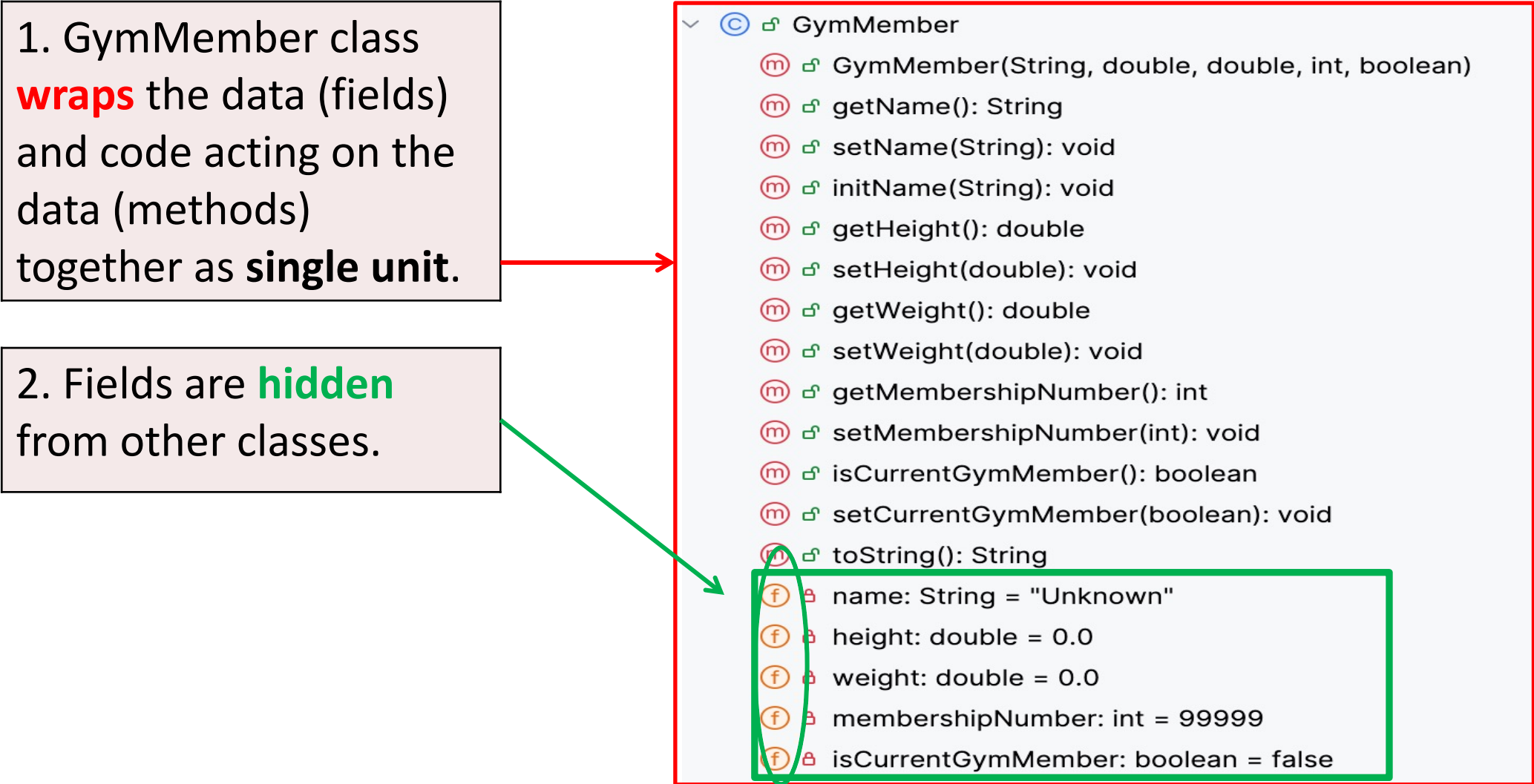


```
▼ c GymMember
  m GymMember(String, double, double, int, boolean)
  m getName(): String
  m setName(String): void
  m getHeight(): double
  m setHeight(double): void
  m getWeight(): double
  m setWeight(double): void
  m getMembershipNumber(): int
  m setMembershipNumber(int): void
  m isCurrentGymMember(): boolean
  m setCurrentGymMember(boolean): void
  m toString(): String ↑Object
  f name: String = "Unknown"
  f height: double = 0.0
  f weight: double = 0.0
  f membershipNumber: int = 99999
  f isCurrentGymMember: boolean = false
```

A GymMember Class... An Encapsulated Class

1. GymMember class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.

2. Fields are **hidden** from other classes.



The diagram illustrates the GymMember class structure. A red box encloses the entire class definition, including methods and fields. A red arrow points from the first text box to the class header. A green arrow points from the second text box to the fields section. The fields are highlighted with a green oval and a green box.

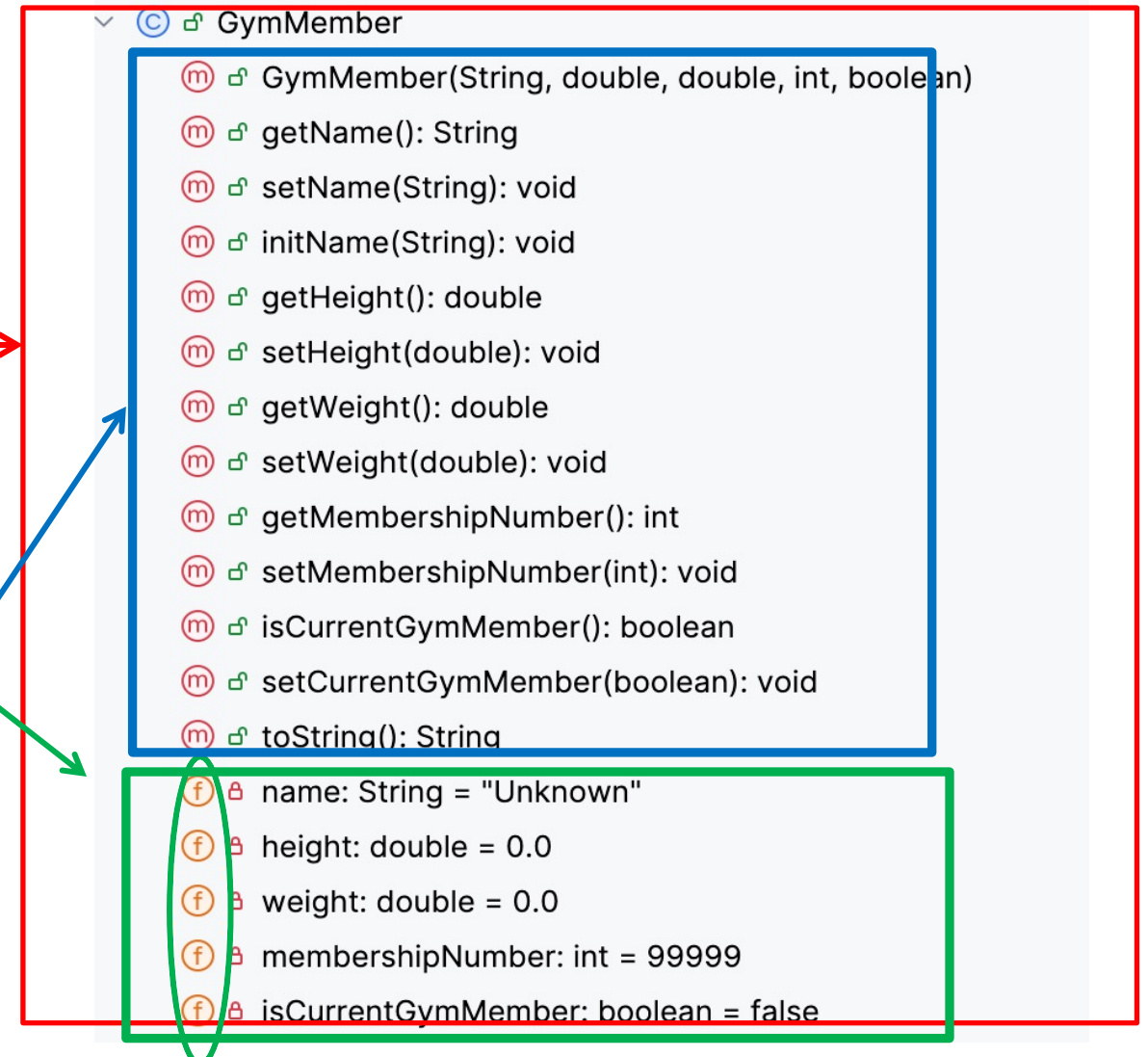
```
✓ GymMember
  (m) GymMember(String, double, double, int, boolean)
  (m) getName(): String
  (m) setName(String): void
  (m) initName(String): void
  (m) getHeight(): double
  (m) setHeight(double): void
  (m) getWeight(): double
  (m) setWeight(double): void
  (m) getMembershipNumber(): int
  (m) setMembershipNumber(int): void
  (m) isCurrentGymMember(): boolean
  (m) setCurrentGymMember(boolean): void
  (m) toString(): String
  (f) name: String = "Unknown"
  (f) height: double = 0.0
  (f) weight: double = 0.0
  (f) membershipNumber: int = 99999
  (f) isCurrentGymMember: boolean = false
```

A GymMember Class... An Encapsulated Class

1. GymMember class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.

2. Fields are **hidden** from other classes.

3. **Access** the fields only through the methods of Product (e.g. **getter** and **setter** methods).





USER I/O

Driver

Using the GymMember Class

1

```
private GymMember gymMember;
```

Declaring an object **gymMember**, of type **GymMember**.

gymMember

null

Using the GymMember Class

1

```
private GymMember gymMember;
```

Declaring an object **gymMember**, of type **GymMember**.

2

```
gymMember = new GymMember("Joe Soap", 1.8, 175, 12001, true);
```

Calls the **GymMember** constructor to build the **gymMember** object in memory.

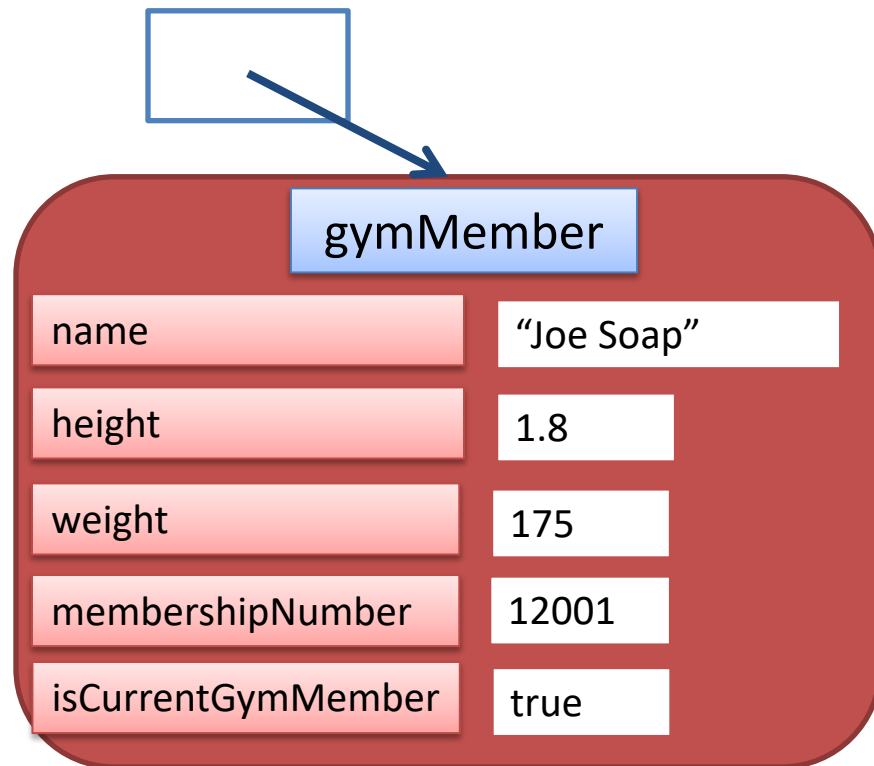
gymMember



Multiple GymMember objects

```
private GymMember gymMember = new GymMember("Joe Soap", 1.8, 175, 12001, true);
```

gymMember

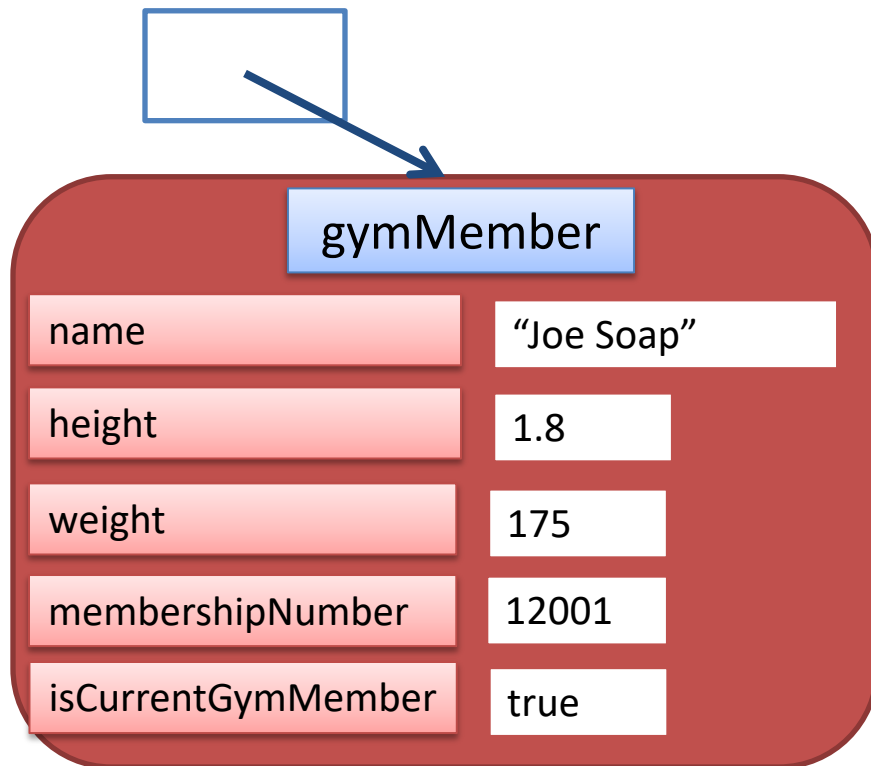


Multiple GymMember objects

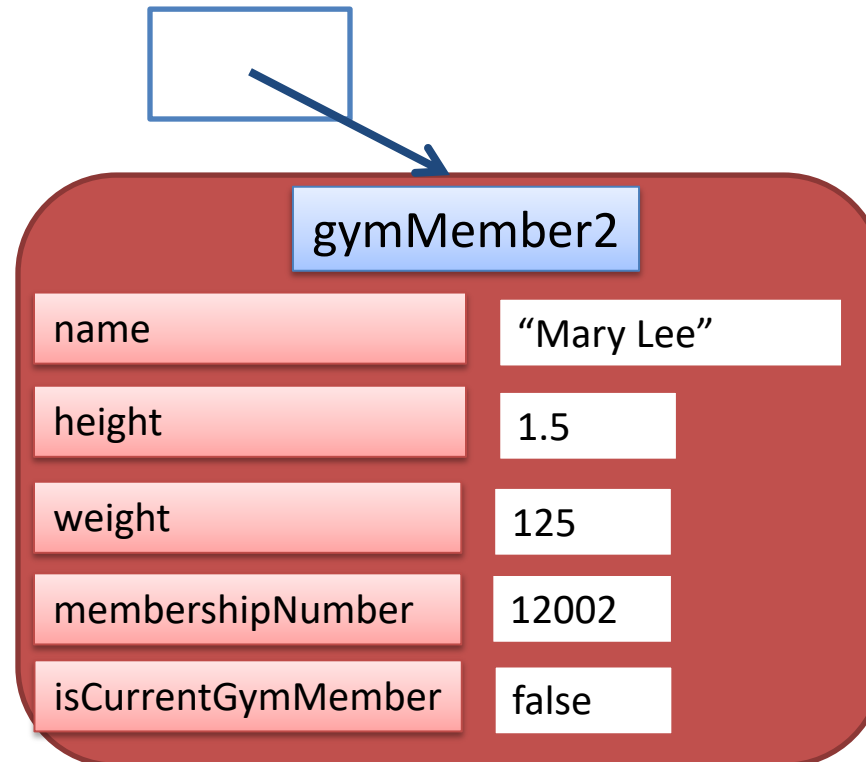
```
private GymMember gymMember = new GymMember("Joe Soap", 1.8, 175, 12001, true);
```

```
private GymMember gymMember2 = new GymMember("Mary Lee", 1.5, 125, 12002, false);
```

gymMember



gymMember2



Driver Class

```
import java.util.Scanner;

public class Driver {

    private Scanner input = new Scanner(System.in);
    private GymMember gymMember;

    public static void main(String[] arg) {
        new Driver();
    }

    public Driver() {
        addGymMember();
        printGymMember();
    }

    // addGymMember () code
    // printGymMember() code

}
```

Driver Class

```
private void addGymMember() {  
    //obtaining the data from the user  
    System.out.println("Entering details");  
    System.out.println("-----");  
    System.out.print("  Enter your name:      ");  
    String name = input.nextLine();  
    System.out.print("  Enter your height (meters): ");  
    double height = input.nextDouble();  
    System.out.print("  Enter your weight (kgs):  ");  
    double weight = input.nextDouble();  
    System.out.print("  Enter the membership number: ");  
    int membershipNumber = input.nextInt();  
    System.out.print("  Is current member (y/n):  ");  
    char isCurrentMemberChar = input.next().charAt(0);  
  
    boolean isCurrentMember = false;  
    if ((isCurrentMemberChar == 'Y') || (isCurrentMemberChar == 'y')) {  
        isCurrentMember = true;  
    }  
  
    gymMember = new GymMember(name, height, weight, membershipNumber, isCurrentMember);  
}
```

Entering details

```
Enter your name:      Joe Soap  
Enter your height (meters): 1.7  
Enter your weight (kgs):  75  
Enter the membership number: 10001  
Is current member (y/n):  y
```

Driver Class

```
private void printGymMember()
{
    //printing out the data to the user
    System.out.println("\n\nPrinting details");
    System.out.println("-----");
    System.out.println(gymMember);
}
```



```
Run: Driver x
/Library/Java/JavaVirtualMachines/jdk-18.0.1.1.jdk/Contents/Home/b
Entering details
-----
Enter your name: Joe Soap
Enter your height (meters): 1.7
Enter your weight (kgs): 75
Enter the membership number: 10001
Is current member (y/n): y

Printing details
-----
Joe Soap: 1.7M, 75.0KG (Member Num: 10001, current member: true)

Process finished with exit code 0
```

Questions?

